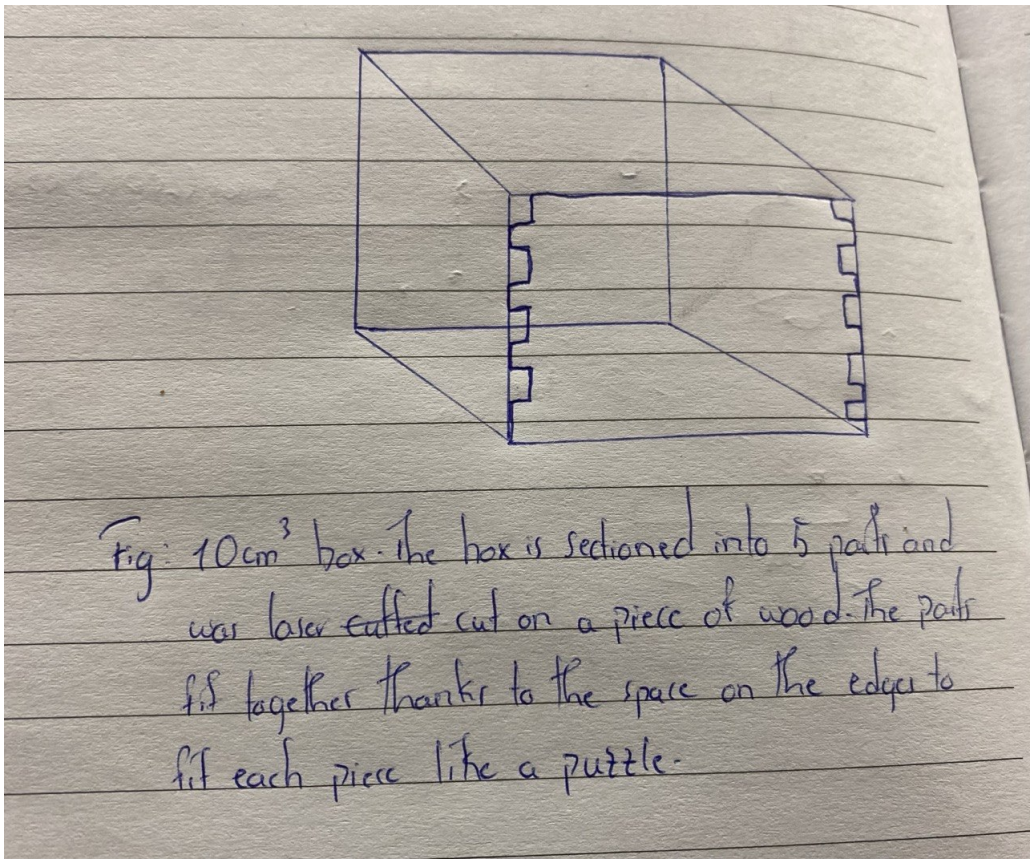
Daily Rant Fab

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Today I was introduced to digital manufacturing. This is the process of creating or manufacturing physical products using software tools and different types of machine. Fusion 360 is a **Computer Aided Design tool** with an integrated **Computer- Aided Manufacturing** (CAM). CAD is the use of computer software to create modify and analyse and optimize 2D and 3D models on physical objects. CAM is the use of software and computer controlled machinery to automate the manufacturing process.

Design Process relies on parametric feature-based modelling, typically beginning with sketches on 2-dimensional planes and utilizing operations such as extrusions, revolutions, sweeps, and more to transition from 2D profiles into complex 3D geometries.

We also had the opportunity to see one of the machines at work namely the Laser cutter. The way it works based on my understanding is through the use of a high hit source (the beam of light or laser) which cuts through the wood based on the design given to it.



Below is the formula to calculate the surface area of a Cube (6 sides).

$$\mathbf{A = 6s^2} \tag{1}$$